

# Suite Hard • Simulation Module 05

22 questions • 35 minutes • official SAT Math Hard

Exclude-active: these items are not on current Bluebook full-length tests.

All items are Hard. This is not an adaptive Module 1. It is harder than test day.

Calculator / Desmos is allowed on every item, same as Bluebook.

Question 1 is the first page after this cover. Write answers on the last page.

\*\*\*Read the prompt carefully.\*\*\* Write the job on scratch before you copy numbers.

Domain mix in this module:

Algebra: 5

Advanced Math: 7

PSDA: 4

Geometry: 6

*Do not open the next module until this one is timed and scored.*

| Assessment | Test | Domain                            | Skill                                                                | Difficulty                                        |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard <div><div></div><div></div><div></div></div> |

Question

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of **85** piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for **75** piano students in this sample who practiced at least **50** minutes on average each week.

|                                               | Percentage of correct notes the students played during a recital |            |                  |       |
|-----------------------------------------------|------------------------------------------------------------------|------------|------------------|-------|
| Average number of minutes practiced each week | Less than 50%                                                    | 50% to 80% | Greater than 80% | Total |
| 50 to 150                                     | 7                                                                | 28         | 5                | 40    |
| 151 or more                                   | 3                                                                | 11         | 21               | 35    |
| Total                                         | 10                                                               | 39         | 26               | 75    |

Which of the following is the largest population to which the results of the study can be generalized?

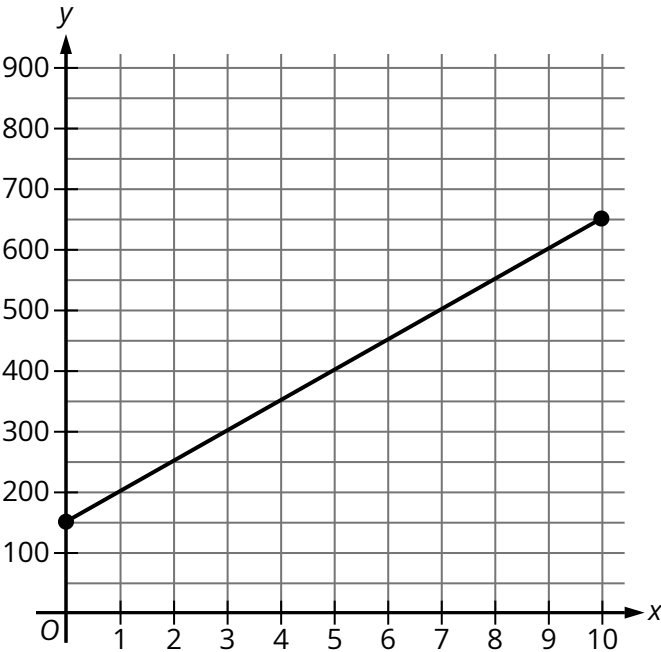
Answer

- A. All piano students who were in their second year with their conservatory
- B. All piano students in the region
- C. All piano students in the region who were in their second year with their conservatory
- D. The **75** piano students in the sample who practiced at least **50** minutes on average each week

| Assessment | Test | Domain  | Skill            | Difficulty                               |
|------------|------|---------|------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard <div></div> <div></div> <div></div> |

**Question**

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance  $y$ , in meters, Mary had walked on the trail  $x$  minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the **10** minutes after leaving the trail sign?

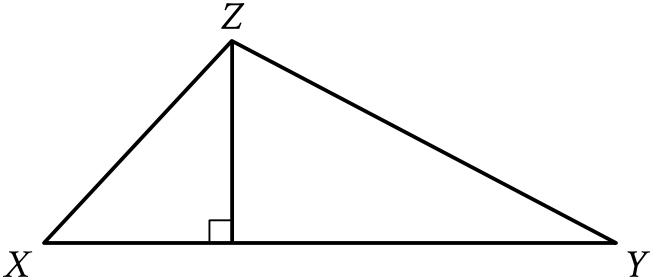
| Assessment | Test | Domain        | Skill                  | Difficulty                                        |
|------------|------|---------------|------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div><div></div><div></div><div></div></div> |

**Question**

The expression  $6x^4 + 31x^2 + 35$  can be rewritten as  $(3x^2 + a)(2x^2 + b)$ , where  $a$  and  $b$  are positive integers, or as  $(3x^2 + c)(2x^2 + d)$ , where  $c$  and  $d$  are positive nonintegers. What is the value of  $a + c$ ?

| Assessment | Test | Domain                    | Skill                            | Difficulty                               |
|------------|------|---------------------------|----------------------------------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Right triangles and trigonometry | Hard <div></div> <div></div> <div></div> |

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle  $X$  is  $54^\circ$ . The length of  $\overline{XY}$  is  $26$  units and the length of  $\overline{XZ}$  is  $19$  units. What is the area, in square units, of triangle  $XYZ$ ?

Answer

- A. ~~247~~
- B. ~~494~~
- C. ~~247~~  $\sin 54^\circ$
- D. ~~494~~  $\sin 54^\circ$

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                        |
|------------|------|-----------------------------------|------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard <div><div></div><div></div><div></div></div> |

Question

On average, a certain plant grows **87** millimeters every ***m*** months. At this rate, which expression represents the number of millimeters, on average, the plant grows every ***k*** years?

Answer

- A.  $\frac{29m}{4k}$
- B.  $\frac{29k}{4m}$
- C.  $\frac{1,044m}{k}$
- D.  $\frac{1,044k}{m}$

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard <div><div></div><div></div><div></div></div> |

**Question**

Which system of linear equations has no solution?

- Answer**
- A.  $-2x + 3y = -9$
- $2x - 3y = 9$
- B.  $2x - 3y = 9$
- $3x + 4y = 10$
- C.  $2x - 3y = 9$
- $-6x + 9y = -27$
- D.  $-2x + 3y = 9$
- $4x - 6y = 18$

| Assessment | Test | Domain        | Skill                  | Difficulty                                        |
|------------|------|---------------|------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div><div></div><div></div><div></div></div> |

Question

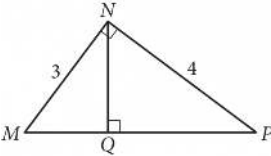
$0.36x^2 + 0.63x + 1.17$

The given expression can be rewritten as  $a(4x^2 + 7x + 13)$ , where  $a$  is a constant. What is the value of  $a$ ?



| Assessment | Test | Domain                    | Skill                        | Difficulty                                        |
|------------|------|---------------------------|------------------------------|---------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Lines, angles, and triangles | Hard <div><div></div><div></div><div></div></div> |

Question



In the figure above, what is the length of  $\overline{NQ}$  ?

Answer

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

| Assessment | Test | Domain                            | Skill       | Difficulty                                        |
|------------|------|-----------------------------------|-------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Percentages | Hard <div><div></div><div></div><div></div></div> |

Question

The number of audiobooks in a library increased by **131%** from last year to this year. The number of audiobooks in the library last year was ***n***, and the number of audiobooks in the library this year is ***bn***, where ***b*** and ***n*** are constants. What is the value of ***b***?

| Assessment | Test | Domain  | Skill                            | Difficulty                                        |
|------------|------|---------|----------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard <div><div></div><div></div><div></div></div> |

Question

$$3(kx + 13) = \frac{48}{17}x + 36$$

In the given equation,  $k$  is a constant. The equation has no solution. What is the value of  $k$ ?

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

Question

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

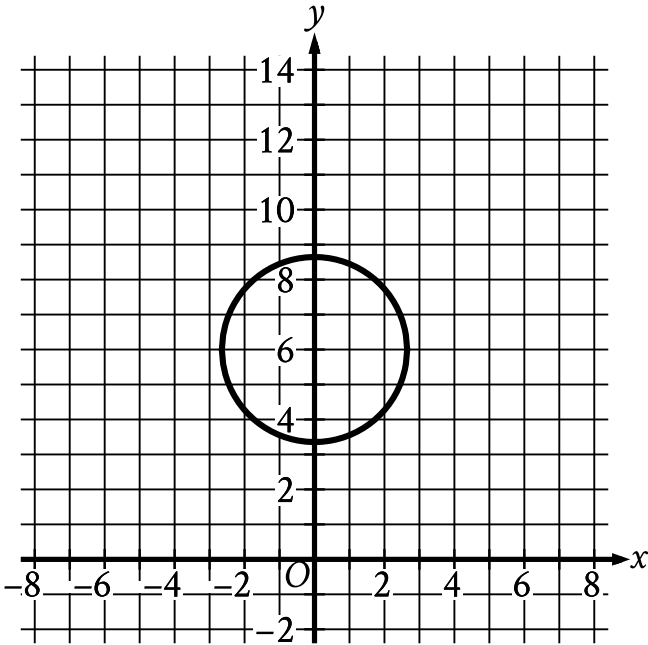
The equation above is true for all  $x > 2$ , where r and t are positive constants. What is the value of rt ?

Answer

- A.  $-20$
- B. 15
- C. 20
- D. 60

| Assessment | Test | Domain                    | Skill   | Difficulty                               |
|------------|------|---------------------------|---------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div></div> <div></div> <div></div> |

Question



Circle A shown is defined by the equation  $x^2 + (y - 6)^2 = 7$ . Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is  $(x - h)^2 + (y - k)^2 = a$ , where  $h$ ,  $k$ , and  $a$  are constants, what is the value of  $4a$ ?

| Assessment | Test | Domain                            | Skill       | Difficulty                                        |
|------------|------|-----------------------------------|-------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Percentages | Hard <div><div></div><div></div><div></div></div> |

Question

Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is  $\frac{3}{4}$  cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If p percent of an adult’s daily allowance of potassium is provided by x servings of Crunchy Grain cereal per day, which of the following expresses p in terms of x ?

- Answer
- A.  $p = 0.5x$
- B.  $p = 5x$
- C.  $p = (0.05)^x$
- D.  $p = (1.05)^x$

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$$-x - wy = -337$$
$$2x - wy = 47$$

In the given system of equations,  $w$  is a constant. In the  $xy$ -plane, the graphs of these equations intersect at the point  $(q, 19)$ , where  $q$  is a constant. What is the value of  $w$ ?

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

**Question**

$h(x) = -16x^2 + 100x + 10$

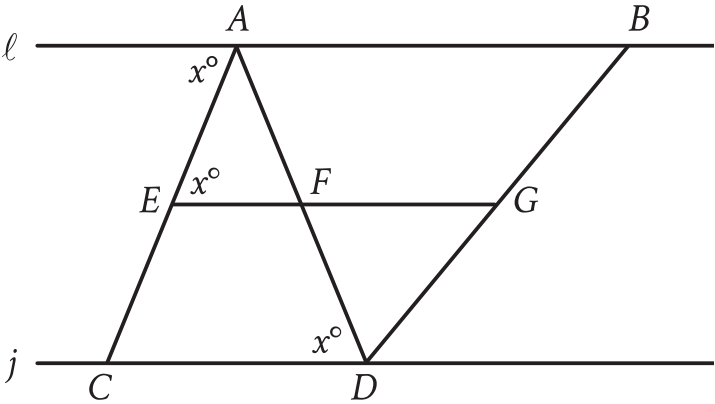
The quadratic function above models the height above the ground h, in feet, of a projectile x seconds after it had been launched vertically. If  $y = h(x)$  is graphed in the xy-plane, which of the following represents the real-life meaning of the positive x-intercept of the graph?

- Answer**
- A. The initial height of the projectile
  - B. The maximum height of the projectile
  - C. The time at which the projectile reaches its maximum height
  - D. The time at which the projectile hits the ground



| Assessment | Test | Domain                    | Skill                        | Difficulty                               |
|------------|------|---------------------------|------------------------------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Lines, angles, and triangles | Hard <div></div> <div></div> <div></div> |

Question



Note: Figure not drawn to scale.

In the figure,  $\overline{EG}$  and  $\overline{AD}$  intersect at point  $F$ , and line **Math output error** is parallel to line  $j$ . If  $EF = 0.5CD$  and  $FG = 2.25CD$ , which of the following must be true?

- I.  $AB = 2.25EF$
- II.  $FG = 0.5AB$
- III.  $\frac{EF}{FG} = \frac{AF}{FD}$

Answer

- A. I only
- B. II only
- C. I and II
- D. II and III

| Assessment | Test | Domain  | Skill                                       | Difficulty                                        |
|------------|------|---------|---------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Linear inequalities in one or two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$y > 2x - 1$

$2x > 5$

Which of the following consists of the y-coordinates of all the points that satisfy the system of inequalities above?

Answer

- A.  $y > 6$
- B.  $y > 4$
- C.  $y > \frac{5}{2}$
- D.  $y > \frac{3}{2}$

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

**Question**

Function  $f$  is defined by  $f(x) = -a^x + b$ , where  $a$  and  $b$  are constants. In the  $xy$ -plane, the graph of  $y = f(x) - 12$  has a  $y$ -intercept at  $(0, -\frac{75}{7})$ . The product of  $a$  and  $b$  is  $\frac{320}{7}$ . What is the value of  $a$ ?

| Assessment | Test | Domain                    | Skill                            | Difficulty                                        |
|------------|------|---------------------------|----------------------------------|---------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Right triangles and trigonometry | Hard <div><div></div><div></div><div></div></div> |

**Question**

Triangle  $ABC$  is similar to triangle  $DEF$ , where angle  $A$  corresponds to angle  $D$  and angle  $C$  corresponds to angle  $F$ . Angles  $C$  and  $F$  are right angles. If  $\tan(A) = \frac{50}{7}$ , what is the value of  $\tan(E)$ ?

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                        |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D, in degrees Fahrenheit, given the temperature T, in degrees Fahrenheit, and the relative humidity of H percent, where  $H > 50$ . Which of the following expresses the relative humidity in terms of the temperature and the dew point?

Answer

- A.  $H = \frac{25}{9}(D - T) + 100$
- B.  $H = \frac{25}{9}(D - T) - 100$
- C.  $H = \frac{25}{9}(D + T) + 100$
- D.  $H = \frac{25}{9}(D + T) - 100$

| Assessment | Test | Domain                    | Skill           | Difficulty                                        |
|------------|------|---------------------------|-----------------|---------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard <div><div></div><div></div><div></div></div> |

Question

The table gives the areas and perimeters of two similar rectangles, where  $n$  is a constant.

|             | Area (square inches) | Perimeter (inches) |
|-------------|----------------------|--------------------|
| Rectangle A | 630                  | 210                |
| Rectangle B | 2,520                | $n$                |

What is the value of  $n$ ?

Answer

- A. 2,100
- B. 1,680
- C. 840
- D. 420

| Assessment | Test | Domain        | Skill               | Difficulty                                        |
|------------|------|---------------|---------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div><div></div><div></div><div></div></div> |

Question

Growth of a Culture of Bacteria

| Day | Number of bacteria per milliliter at end of day |
|-----|-------------------------------------------------|
| 1   | $2.5 \times 10^5$                               |
| 2   | $5.0 \times 10^5$                               |
| 3   | $1.0 \times 10^6$                               |

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach  $5.12 \times 10^8$ ?

- Answer
- A. Day 5

B. Day 9

C. Day 11

D. Day 12

## Module 05 • answer sheet

Write one answer per line. SPR: integer, decimal, or fraction.

1. \_\_\_\_\_

2. \_\_\_\_\_

3. \_\_\_\_\_

4. \_\_\_\_\_

5. \_\_\_\_\_

6. \_\_\_\_\_

7. \_\_\_\_\_

8. \_\_\_\_\_

9. \_\_\_\_\_

10. \_\_\_\_\_

11. \_\_\_\_\_

12. \_\_\_\_\_

13. \_\_\_\_\_

14. \_\_\_\_\_

15. \_\_\_\_\_

16. \_\_\_\_\_

17. \_\_\_\_\_

18. \_\_\_\_\_

19. \_\_\_\_\_

20. \_\_\_\_\_

21. \_\_\_\_\_

22. \_\_\_\_\_